Ask a Language Model for Lottery Numbers

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Abstract

A mechanical lottery draw uses effectively 41 of 49 numbers in each of two summers 27 years apart and sits inside the fair-sampling envelope, while six current language models asked 200 times each for six random numbers use effectively 10 to 18 of the 49 and fall far outside it, the most frequent ticket of every model opens on 7, and one model returns a single ticket in half of 200 asks.

1 An audited machine as the null

Sampling temperature makes a language model's output vary from run to run, and that variation is easy to read as randomness. This note measures the variation against the strongest available null: a mechanical, audited lottery draw. Polish Lotto publishes its draw statistics, and two summers of the game, 27 years apart, give an empirical picture of what a genuinely random source looks like at small sample sizes, with no argument about what "random enough" means in theory.

Every source is scored by the effective number of numbers it uses, $N_{\text{eff}} = 1/\sum_{i=1}^{49} s_i^2$, where s_i is the share of all returned balls landing on number i. A source using all 49 numbers equally scores 49; a source always returning the same six scores 6. A fair source at a small sample is lumpy, so we simulate 20,000 fair sources at each source's exact ticket count, six distinct numbers per ticket, and call a source outside the fair envelope if its N_{eff} falls below the simulated 5th percentile or its top number's count exceeds the 95th percentile. The method and the expected outcomes were registered before any model call (`SPEC.md` and `PREDICTION.md` in the data package).

Figure 1 carries the result, and Table 1 the numbers. The machine used effectively 41.1 of 49 numbers in summer 1999 and 41.4 in summer 2026, inside its envelope both times. The six models, at five times the machine's sample size, used 9.9 to 18.0 against a fair floor of 46.7 at 1,200 balls; all six are outside, and none is close.

2 Experiment and results

Design. Six engines (gpt-5.6-luna, claude-sonnet-5, gemini-3.7-flash, grok-4.5, mistral-large, Perplexity sonar) were each asked 200 times, in English, for six different random numbers between 1 and 49, with four prompt paraphrases rotated across runs. Every call opened a fresh context. Temperature was the provider default, never 0: at temperature 0 one measures the mode, and

Figure 1: Share of balls per number, 1 to 49, on a shared y-scale. Top row: Polish Lotto, summers 1999 and 2026, the operator's published draw statistics. Rows two to four: six engines, 200 runs each, four rotated paraphrases, provider-default temperature, fresh context per call, web search off except sonar, 1 September 2026. Mint bars mark the six numbers of each engine's most frequent ticket; the machine has no most frequent anything.

Table 1: Spread of returned numbers per source. The fair floor is the 5th percentile of N_{eff} over 20,000 simulated fair sources at that source's ticket count, six distinct numbers per ticket; p is chi-square against uniform, $df = 48$.

source	balls	N_{eff} of 49	fair floor	top number	p	envelope
Polish Lotto, summer 1999	156	41.1	35.7	4.5%	0.98	inside
Polish Lotto, summer 2026	234	41.4	39.3	4.3%	0.68	inside
grok-4.5	1,200	18.0	46.7	9.6%	< 0.0001	outside
claude-sonnet-5	1,176	13.5	46.7	14.4%	< 0.0001	outside
mistral-large	1,128	13.2	46.6	14.3%	< 0.0001	outside
gpt-5.6-luna	1,200	12.4	46.7	8.8%	< 0.0001	outside
sonar (Perplexity)	1,200	10.5	46.7	14.6%	< 0.0001	outside
gemini-3.7-flash	1,200	9.9	46.7	14.6%	< 0.0001	outside

Table 2: Whole-ticket behaviour per engine. Valid runs are the parsed answers of 200 asks; the modal share is over valid runs.

engine	valid runs	distinct tickets	modal ticket	modal share
gpt-5.6-luna	200	8	7 14 22 31 38 46	50.5%
mistral-large	188	24	7 12 19 23 31 45	26.6%
sonar (Perplexity)	200	30	7 14 23 28 35 46	68.0%
claude-sonnet-5	196	36	7 14 22 29 35 41	26.0%
gemini-3.7-flash	200	36	7 14 23 31 42 48	38.0%
grok-4.5	200	93	7 14 23 31 42 48	22.5%

the distribution is the point. Web search was off, except sonar, which searches by design. Cells ran in randomised order on 1 September 2026. The baseline is the operator's published draw statistics for 1 June to 31 August 1999 (26 draws, 156 balls) and the same window in 2026 (39 draws, 234 balls). Chi-square against uniform rejects every engine at $p < 0.0001$; the machine's two summers come out at $p = 0.98$ and $p = 0.68$.

Tickets. The stronger result sits at the level of whole answers (Table 2). There are 13,983,816 possible tickets, so a fair source asked 200 times repeats any ticket at all with probability about one in 700. gpt-5.6-luna produced 8 distinct tickets in 200 asks and returned its most frequent one, 7 14 22 31 38 46, in 101 of them. sonar gave the same six numbers in 68% of all answers. Every one of the six modal tickets opens on 7: six models, five companies, one first number. gemini-3.7-flash and grok-4.5, from different vendors, share the identical modal ticket, 7 14 23 31 42 48.

The number 42 splits the field. gemini-3.7-flash, grok-4.5 and mistral-large rank it fifth of 49 (7.5%, 8.3% and 7.2% of balls) and claude-sonnet-5 eighth (4.8%); sonar ranks it 30th (0.2%), and gpt-5.6-luna never returned it once in 1,200 balls. In the machine's two summers 42 was drawn once each and ranked 48th and 49th of 49, which is what chance does to some number at these sample sizes; the machine holds no view on 42, and neither ranking supports a story about it.

Where the pre-registered prediction failed. Two design drifts first, disclosed plainly: the prediction document named four engines and gemini-3.6-flash; the run covered six engines and gemini-3.7-flash, so the prediction binds the four it named and reads as directional for the other two. The prediction put every engine below the fair floor at N_{eff} 20 to 35; the observed 9.9 to 18.0 is a stronger collapse than predicted. The predicted human birthday bias is absent: the prediction expected 68 to 78% of balls in the range 1 to 31 against the fair 63.3%, and the models came in at 63.8 to 68.0%, the same band as the machine's own 67.3% and 68.8%. Mean ball ran 24.4 to 27.0 against a predicted drift below 25.0. The models did not inherit the human habit of marking birthdays; their defect is collapse onto a few remembered tickets. Ordering behaved as predicted and then some: gemini and gpt-5.6-luna returned the six numbers already sorted in 100% of parsed answers, claude in 99.5%, sonar in 99.0%, grok in 87.0%, and mistral in 27.7%. A machine draws in no order; a source that nearly always sorts is filling a template.

Data quality. Of 1,200 calls, 1,188 were answered; all 12 failures are HTTP 429 rate limits on mistral-large during the parallel run, a collection artifact. Four claude-sonnet-5 answers (2.0%) did not parse to six numbers and are excluded, with the verbatim text kept. No parsed answer contained a duplicate or an out-of-range number.

3 Related work

Non-determinism is routinely mistaken for randomness: sampling temperature makes outputs vary, and this experiment shows the variation is concentrated, not uniform. That identical prompts yield different answers is established [1], and our earlier work decomposes the variance of repeated brand answers [4] and formalizes repeated-query audit protocols [5]. LLMs are known to produce biased binary sequences when asked to flip coins [2] and biased numbers across ranges, temperatures and prompt languages [3]; both lines score models against a theoretical target distribution. This note adds two things: an audited physical device as an empirical null at matched sample sizes, and the ticket level, where the collapse is strongest, with whole six-number answers recurring and two rival engines sharing one.

4 Conclusion

Asked the same question 200 times, the machine gave a fresh answer every time and the models reached for answers they had reached for before. For anyone sampling model output, run-to-run variation is no evidence of coverage: every engine put between 22% and 68% of its answers on one ticket.

Future directions. (1) Other languages: the collector's Polish arm (`pl_base`) is built and unrun, and favourite numbers may be locale-specific. (2) Temperature sweeps: how N_{eff} moves with decoding temperature. (3) Fixed seeds: whether seeded sampling collapses the ticket set further. (4) The same design on commercial questions, where the collapsed set is brands; our companion note measures that repertoire [6]. (5) The number-theoretic structure of the favourites: the modal tickets step across the wheel in near-even gaps, which invites a formal description. (6) Comparison with the human favourite-number literature: the models dodge the human birthday bias yet still open on 7.

Declarations

Competing interests. The author is CEO of Rankfor.AI, which sells AI-visibility measurement; this note is methodological and names no customer. **Pre-registration.** The method (`SPEC.md`) and the expected outcomes (`PREDICTION.md`) were written before any model call was made. **Data and code.** All raw answers, per-number counts, metrics, the collector, and both pre-registration files will be mirrored at `https://github.com/Rankfor/rankfor-open`; they currently live in the author's operations repository under `rankfor-ops/scripts/lotto-vs-models`. **Provenance.** The Lotto figures are the operator's published draw statistics for 1 June to 31 August 1999 and 2026; they are the operator's collection.

References

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[5] Żatuchin D (2026) The Dice Roll Method: a standardized protocol for repeated-query auditing of large language model brand recommendations. Manuscript; validation pack at github.com/Rankfor/rankfor-open

[6] Żatuchin D (2026) Repeated queries exhaust an LLM's brand recommendations but not its sources. Companion note, in the same repository